Defuzzification Methods

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Lambda-cut method

- The lambda cut method of defuzzification converts a fuzzy set into a crisp (non-fuzzy) set by selecting all elements whose **membership value is greater than or equal to a specific threshold, called lambda(λ).**
- In other-words, $A_\lambda = \{x | \mu A(x) \geq \lambda\}$
- This Lambda-cut set A_λ is also called alpha-cut set.
- A1 = {(x1, 0.9), (x2, 0.5), (x3, 0.2), (x4, 0.3)}
- Then $A_{0.6}$ = {(x1, 1), (x2, 0), (x3, 0), (x4, 0)} = {x1}

Two fuzzy sets P and Q are defined on x as follows.

$\mu(x)$	x_1	x_2	x_3	x_4	x_5
P	0.1	0.2	0.7	0.5	0.4
Q	0.9	0.6	0.3	0.2	0.8

Find the following :

(a) $P_{0.2}$, $Q_{0.3}$

(b) $(P \cup Q)_{0.6}$

(c) $(P \cup \overline{P})_{0.8}$

(d) $(P \cap Q)_{0.4}$

The Lambda-cut method for a fuzzy set can also be extended to fuzzy relation also.

Example: For a fuzzy relation R

$$R = \begin{bmatrix} 1 & 0.2 & 0.3 \\ 0.5 & 0.9 & 0.6 \\ 0.4 & 0.8 & 0.7 \end{bmatrix}$$

We are to find λ-cut relations for the following values of $\lambda = 0, 0.2, 0.9, 0.5$

$$R_0 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \text{ and } R_{0.2} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \text{ and}$$

$$R_{0.9} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } R_{0.5} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

- **Maxima Methods**
  1. Height method
  2. First of maxima (FoM)
  3. Last of maxima (LoM)
  4. Mean of maxima(MoM)
- **Centroid methods**
  1. Center of gravity method (CoG)
  2. Center of sum method (CoS)
  3. Center of area method (CoA)
- **Weighted average method**

Maxima Method: Max- Membership Method

- A category of methods used in fuzzy logic to convert a fuzzy set into a single, precise ("crisp") output value **by selecting the element(s) with the highest degree of membership.**
- **First of Maxima (FOM):** This method selects the **smallest** value in the domain that has the maximum membership degree.
- **Last of Maxima (LOM):** This method selects the **largest** value in the domain that has the maximum membership degree.
- **Mean of Maxima (MOM):** This method calculates the **average (mean)** of all the values in the domain that share the maximum membership degree. This is commonly known as the middle-of-maxima method.

Suppose, a fuzzy set **Young** is defined as follows:

Young = $\{(15,0.5), (20,0.8), (25,0.8), (30,0.5), (35,0.3)\}$

Then the crisp value of **Young** using MoM method is

$x^* = \frac{20+25}{2} = 22.5$

Thus, a person of 22.5 years old is treated as young!

Centroid Method

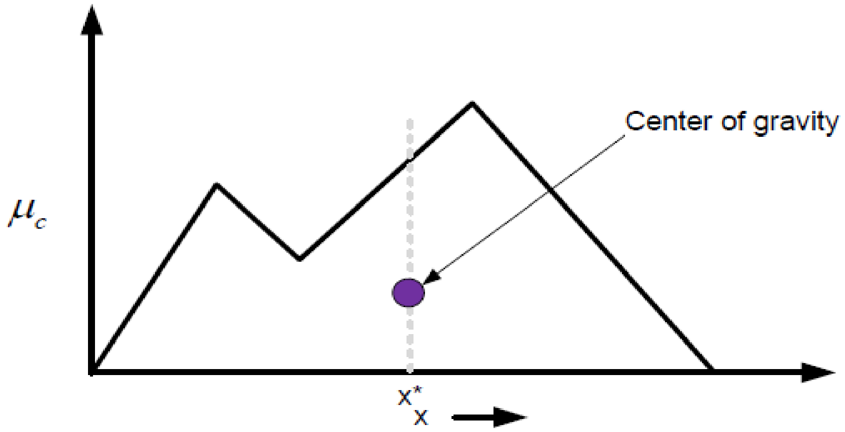
- **Center of Gravity(COG)**

1. The basic principle in CoG method is to find the point x^* where a vertical line would slice the aggregate into two equal masses.

2. Mathematically, the CoG can be expressed as follows :

$$x^* = \frac{\int x.\mu_C(x)dx}{\int \mu_C(x)dx}$$

3. Graphically,



Note:

1. x^* is the x-coordinate of center of gravity.

2. $\int \mu_C(x)dx$ denotes the area of the region bounded by the curve μ_C.

3. If μ_C is defined with a discrete membership function, then CoG can be stated as :

   $$x^* = \frac{\sum_{i=1}^{n} x_i.\mu_C(x_i)}{\sum_{i=1}^{n} \mu_C(x_i)} \;;$$

4. Here, x_i is a sample element and n represents the number of samples in fuzzy set C.

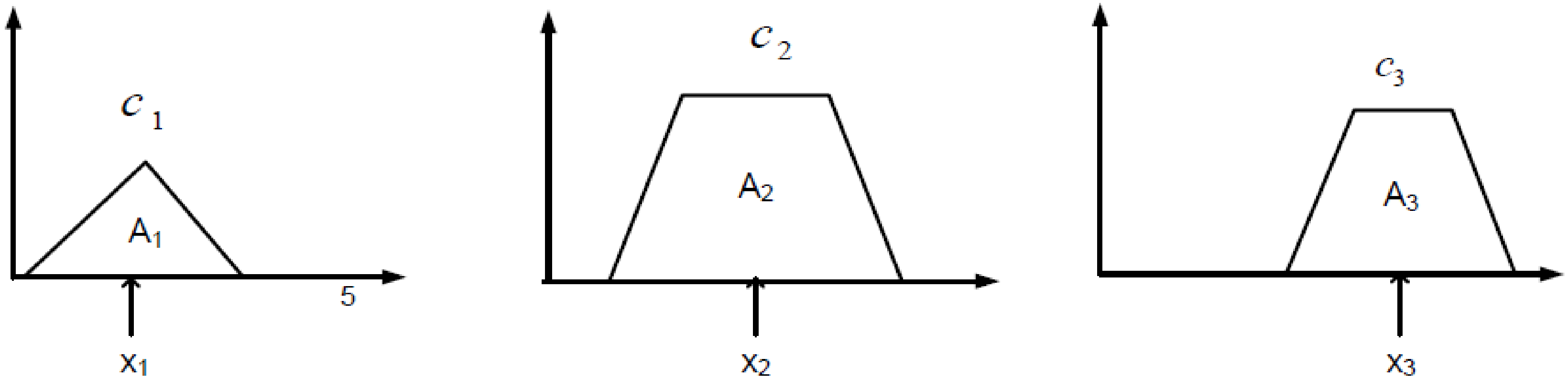
- **Center of Sum (COS)**

If the output fuzzy set $C = C_1 \cup C_2 \cup ....C_n$, then the crisp value according to CoS is defined as

$$x^* = \frac{\sum_{i=1}^{n} x_i . A_{c_i}}{\sum_{i=1}^{n} A_{c_i}}$$

Here, A_{c_i} denotes the area of the region bounded by the fuzzy set C_i and x_i is the geometric center of the area A_{c_i}.

Graphically,



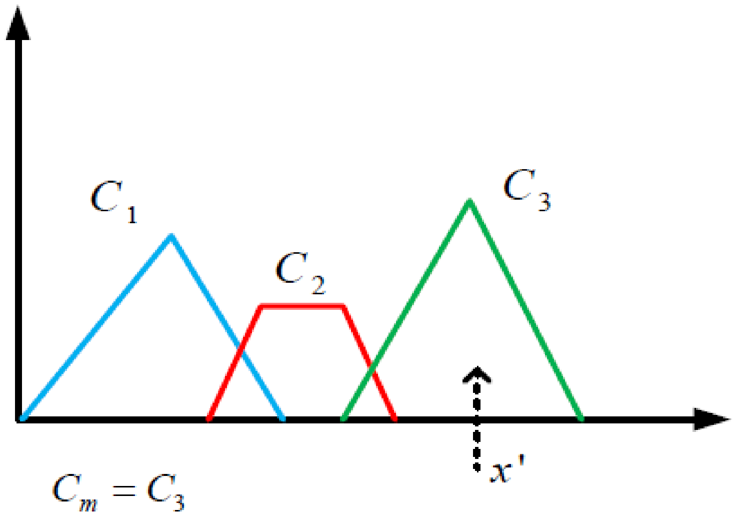
- **Center of largest Area (COA)**

  If the fuzzy set has two subregions, then the center of gravity of the subregion with the largest area can be used to calculate the defuzzified value.

  Mathematically, $x^* = \frac{\int \mu_{cm}(x).x' dx}{\int \mu_{cm}(x) dx}$;

  Here, C_m is the region with largest area, x' is the center of gravity of C_m.

  Graphically,



Weighted Average Method

- The weighted average (WA) defuzzification method finds a crisp output by calculating the weighted average of the centers of the output fuzzy sets, where each set is weighted by its maximum membership value.

$$x^* = \frac{\sum(\mu_{max_i} \times C_i)}{\sum \mu_{max_i}}$$

  - x^* is the crisp defuzzified output.
  - μ_{max_i} is the maximum membership value of the i-th fuzzy set.
  - C_i is the center of the i-th fuzzy set (the crisp value where its membership is maximal).

Example

Given

- **Fuzzy Set 1**:
    - Center (C_1): 3
    - Maximum Membership (μ_{max1}): 0.7
- **Fuzzy Set 2**:
    - Center (C_2): 7
    - Maximum Membership (μ_{max2}): 1.0

$$x^* = \frac{(0.7 \times 3) + (1.0 \times 7)}{0.7 + 1.0} = \frac{2.1 + 7}{1.7} = \frac{9.1}{1.7} \approx 5.35$$